



Sejong University, Seoul, South Korea

OPERATING SYSTEM

Assignment 2

Processes & System Calls

Time Allowed: 1 Week **Deadline is 12 April 2026 (by 11:59 PM)**

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Learning Objectives

- Distinguish between a **program** and a **process**, and explain how the OS manages processes.
- Describe **kernel mode vs. user mode** and the role of **system calls (trap)**.
- Write and analyze C programs using `fork()`, `exec()`, and `getpid()`.
- Inspect live processes with Linux tools: `ps`, `top`, `strace`, `/proc`.

Part-I: Process Fundamentals

- 1) What is the difference between a program and a process? Can multiple processes be created from a single program? Give a real-world example.
- 2) What information does the OS store in a Process Control Block (PCB)? Why is the PCB essential when performing a context switch?

Part-2: System Calls & Kernel/User Mode

- 3) Explain the difference between **User Mode** and **Kernel Mode**. Why can user programs not directly access hardware devices?
- 4) Describe step-by-step what happens when a user program calls `printf()` until text appears on screen. Your answer must mention: GLIBC, the `write` syscall, the **trap instruction**, privilege escalation, and **return-from-trap**.

Part-3: Exploring Processes with Linux (Hands-on)

Run each command, provide the output, and write one sentence explaining what it shows:

- 5) Search different 20 commands used for process ps. Show the output your own shell process.
Example: `ps -el`

C Programming: fork()

Write, compile, and run the program below. Answer the questions that follow:

Practice-1: What you See how may process are there.

```
#include <stdio.h>
#include <sys/types.h>
#include <unistd.h>
int main() {
    fork();
    printf("Hello from user space!\n PID = %d\n", getpid());
    return 0;
}
```

Run the Code

```
gcc fork_demo.c -o fork_demo && ./fork_demo
```

Practice-2: What you See how may process are there. Why there are many process, explain reason and formula

```
#include <stdio.h>
#include <sys/types.h>
#include <unistd.h>
int main() {
    fork();
    fork();
    fork();
    printf("Hello from user space!\n PID = %d\n", getpid());
    return 0;
}
```

Run the Code

```
gcc fork_demo.c -o fork_demo && ./fork_demo
```

Submission Guidelines

What to Submit

- **Written answers (Parts 1 & 2):** Single PDF all conceptual answers and terminal outputs/screenshots.
- **Screenshots (Parts 3):** Clearly labelled screenshots for every hands-on task in Part 3.

File Naming & Delivery

- Make a single PDF file (Only PDF Acceptable): `StudentID_Name_OS_Assignment2.pdf`
- Submit via course portal
- Late submissions are not allowed.

Academic Integrity Warning

All work must be your own. Code must compile and run correctly on Ubuntu Linux. You may discuss concepts with peers, but written answers and code must be written individually. Plagiarised submissions will receive zero marks and be reported to the department.